

Supplementary material for PLECTA: Overlap-aware instance reconstruction of dense filament networks from binary masks

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S1 Measures

Every measure this paper reports is defined here, with what it rewards and its blind spot; none reads the rendered width of an instance.

The common-fragment set, shared by the first five measures below. The mask is skeletonised, spurs of at most 3 px are pruned, skeleton pixels with three or more 8-neighbours are deleted, the remaining 8-connected pieces are labelled, and pieces shorter than 6 px are discarded, so one fragment is one unambiguous arm between crossings. The partition reads neither the reference nor any prediction, so every method compared is cut into the same fragments. Each then takes a reference and a predicted identity by one rule, applied to both sides alike: the fragment is assigned to the instance whose mask overlaps the largest number of its pixels, the vote counted against each instance’s full mask rather than a flattened label image, so an instance that overlaps another keeps its pixels, provided that the winning overlap covers at least 30 % of the fragment; failing that, it falls back to the most frequent nearest instance label within 6 px of the fragment, read from a deterministically flattened field; failing both, it stays unassigned and is scored as its own singleton cluster, on the reference side and the prediction side equally, so every fragment enters the scored partition. Ties at any step resolve to the lowest instance id, and the flattening order is fixed, so the assignment is deterministic. Because junction pixels are deleted first, no fragment contains one: none of these measures can see whether an emitted crossing pixel is given one owner or two, only whether the arms meeting there are grouped as the reference groups them.

Sensitivity to the scorer’s constants. Because the four constants above are shared with the method (the article’s metrics section), the stored predictions of all four methods were re-scored under one-at-a-time perturbations of each constant, with the grid frozen in writing before any result was computed: spur pruning 2–4 px, minimum fragment length 4–8 px, overlap threshold 0.20–0.50, fallback distance 3–9 px. Nothing was re-run and no configuration was changed in response. Table S1 reports PLECTA’s mean F_1 and its plain paired margins over each comparator at every perturbation, over the 50 degraded comparison scenes. Across the 10 combinations PLECTA’s mean F_1 spans 0.713–0.859, and the smallest paired margin over any comparator under any perturbation is 0.115. The level moves only with the minimum fragment length, and in the same direction for every method, because admitting shorter fragments adds hard, low-evidence units for everyone; the other three constants move the mean by less than 0.002, and no perturbation changes the ordering of the four methods.

Common-fragment pairwise F_1 , the primary endpoint. A pair of fragments is a true positive when it shares both identities, a false positive when the prediction joins fragments of different reference filaments, a false negative when it separates fragments of one:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}. \quad (S1)$$

A clustering measure of the Rand family [1–3], permutation-invariant in the instance labels; it rewards holding a filament together across every crossing and gap it passes. Blind spots: it is

Table S1. Scorer-constant sensitivity. Stored predictions re-scored under one-at-a-time perturbations of the four common-fragment constants; Δ columns are plain paired mean margins, PLECTA minus comparator, over the scenes both methods completed. Sensitivity reporting only, no inference, and the published constants were not changed in response.

Scorer constants	PLECTA F_1	Δ greedy	Δ DNAi	Δ GraFT
published constants	0.822	0.145	0.478	0.339
spur prune 2 px	0.822	0.145	0.478	0.339
spur prune 4 px	0.822	0.145	0.478	0.339
min fragment 4 px	0.713	0.115	0.388	0.267
min fragment 8 px	0.859	0.149	0.507	0.358
overlap threshold 0.20	0.822	0.145	0.478	0.339
overlap threshold 0.40	0.822	0.145	0.478	0.339
overlap threshold 0.50	0.822	0.145	0.478	0.339
fallback distance 3 px	0.824	0.145	0.478	0.340
fallback distance 9 px	0.821	0.145	0.478	0.340

quadratic in fragmentation, so a filament arriving in n pieces carries $\binom{n}{2}$ pairs and a broken mask is charged many times for one missed join, and it weights long, much-crossed filaments far above short ones.

Adjusted Rand index [4] counts the same pairs corrected for chance, which matters because with a hundred filaments per scene almost every pair genuinely belongs to different filaments and the uncorrected index sits near 1 for anything. Blind spot: redundancy — over the comparison set it never departs from pairwise F_1 by more than 0.019, so it confirms rather than adds.

Variation of information [5], in bits, lower better, is the conditional-entropy distance between the partitions: $VI_{\text{split}} = H(\text{prediction} \mid \text{reference})$ is paid for cutting one filament into several, $VI_{\text{merge}} = H(\text{reference} \mid \text{prediction})$ for fusing several into one, and reporting both is what separates a method that divides from one that fuses. Blind spot: each component alone is degenerate — a method merging everything never splits anything and scores a perfect zero on the split component — so neither is ever quoted singly.

Detection F_1 , the measure DNAi reports, matches predicted to reference *objects* greedily by highest intersection over union, stopping below 0.1, DNAi’s own shipped threshold, with precision over predicted and recall over reference objects. It uses a symmetric 2px placement tolerance, under which a reference pixel counts as found when the prediction passes within it and vice versa, the intersection being the smaller of the two counts. The tolerance is necessary: two independently read one-pixel centrelines two pixels apart intersect in *nothing*, so a strict overlap reports placement rather than shape, a problem the thin-structure benchmark FISBe states as a design rationale [6]. Counting objects makes it the most forgiving reading of a fragmented mask and the least discriminative measure here. Blind spot: an arm swap, which it scores 1.000 where the pairwise score gives 0.000.

Crossing Fidelity states the contribution directly. At every junction node the reference induces a partition of the incident arms — two together exactly when the reference gives them the same filament — and so does the prediction; Crossing Fidelity is the fraction of crossings at which the two partitions are identical. Nodes are the junction clusters of the same pruned skeleton and a node’s arms are the fragments 8-adjacent to it. Only nodes whose arms carry at least two distinct reference filaments are scored, since a node where every arm belongs to one filament is a skeletonisation artefact and scoring it would reward doing nothing. It is an in-house diagnostic, not an established metric. Blind spots, both of which govern its use: it is far less discriminative than the pairwise score, so it says what was solved and is never a ranking axis; and crossings carry no information about *gaps*, so it is silent on the cost the pairwise score pays for bridging them. Sharing its skeleton and assignment with the pairwise score makes it

a sharper view of the same evidence rather than independent confirmation, and it is always printed beside its all-unpaired baseline.

Join F_1 counts decisions rather than pairs, for the real fields where the two mask sources fragment differently. A *decision* is a pair of distinct fragments whose endpoints come within 85 px, the gap this method will bridge; reference and prediction each say join or not, and precision, recall and their harmonic mean are taken over decisions. Being linear rather than quadratic in fragmentation, and conditioned on the mask, it asks two mask sources comparably hard questions, and splitting and merging appear separately as low recall and low precision. Blind spot: a fragment pair the mask never brought within reach counts against nobody, so it cannot see a filament the mask lost entirely, and it is always quoted with the number of decisions each mask posed.

Reference-instance recovery is the fraction of reference filaments recovered whole by one predicted instance. The dominant predicted instance of a filament owns most of its fragments; coverage is that count over the filament’s fragment count, purity the same count over every fragment that instance owns anywhere, and a filament counts as recovered when the dominance is *mutual* and both reach 0.8. Purity is what stops a method that merges everything from scoring 1.0, and the measure weights every filament equally, complementing the pairwise score’s quadratic weighting. Blind spot: the two thresholds, which make it insensitive to how badly a failing filament was reconstructed.

Completeness, correctness and quality characterise the input *mask*: no reconstruction is opened to compute them and no method can change them. Reference axis and test mask are both skeletonised and each tested against the other dilated by the tolerance. Completeness is tolerant recall of the reference axis, correctness tolerant precision of the test axis, and quality the tolerant Jaccard index $TP/(TP + FP + FN)$ with TP the smaller of the two matched counts, so quality cannot exceed either component. Quality is also GraFT’s published Jaccard index in its intended reading. Blind spot: the tolerance carries the measure — at zero tolerance it reports raster placement rather than geometry — so it is never quoted without one.

Shared-pixel precision and recall are precision and recall over the pixels the reference gives to more than one instance, computed on the raw instance masks. They exist because the pairwise measure discards junction pixels and cannot see whether a crossing pixel was marked as jointly owned at all: only a method emitting overlapping layers can score above zero on the recall, and one emitting a hard partition scores exactly zero by construction. Blind spot: neither component distinguishes a pixel shared for the right reason from one shared because a chain was painted with a whole junction cluster, which is the over-marking the article’s crossing-representation results report.

The runtime scaling exponent is the slope of an ordinary least-squares fit of log seconds on log foreground pixels, so an exponent b means cost grows as the b -th power of the amount of ink. It is reported instead of absolute time because these are independently optimised codebases, where absolute time measures implementation as much as algorithm. Blind spot: it describes a median trend and is silent on the tail, so it is quoted beside the per-density medians and the censored-scene count; censoring biases it downwards for any method whose slowest scenes did not finish, which is why GraFT’s is a lower bound.

Two measures were computed and are not endpoints. GraFT’s *filament matched coverage* asks whether each reference filament was matched one-to-one to some detection, and PLECTA ranks *below* both comparators on it: at the densest stratum of GraFT’s own generator, 0.798 against 0.998 and 0.900. It is not an endpoint because it counts existence rather than correctness — a method emitting more objects than the scene contains has a spare for every filament, each method’s matched coverage equals the smaller of one and its emission ratio to within 0.16, and requiring a detection to cover half the axis it claims restores the order. *Centreline Dice* [7] was tried as an alternative matching criterion and not adopted: it tests each side’s skeleton against the other’s *volume*, and a centreline grouping has none, so it degenerates

into the placement test the tolerance exists to correct. No cIDice result is reported in this paper.

S2 Comparator configuration

The article’s comparator section states each adaptation and its cost; this is the mechanism behind it, for a reader checking that the comparison was run fairly. **DNAi** clusters junction pixels by single linkage and erases a disk at each cluster. At its released 10 px linkage distance, crossings closer than that chain into one cluster, which at 60 % coverage reaches 89 px across and is replaced by a single small erasure disk, leaving a whole tangle as one instance, which is why its shipped default is not a fair measure of it. Reducing the distance to 2 px on development scenes raised development F_1 and cut variation-of-information merge by about a factor of three, confirming the mechanism; the tuned configuration is the one reported. Two structural choices account for the residual difference: it commits to a pairing at every junction of degree two to four with no cost threshold, so it cannot decline a continuation the way a priced unmatched option can, and it attempts no junction above degree four, resolving 0.016 of the crossings above it. Those are about a tenth of the crossings, but they carry 37.2 % of same-filament fragment pairs on the degraded masks, one such junction severing every pair that spans it. **GraFT**’s errors are divisions rather than merges (variation-of-information split 1.166 against merge 0.392), which follows from a greedy longest-first path cover that cannot revise an early commitment once the scene collapses into one connected graph, as it does above 30 % coverage.

S3 Depth measures

The measures of Section S1 score which centreline fragments belong to one instance. None of them can see depth: two reconstructions that agree on every instance but disagree about which filament passes in front at every crossing are indistinguishable to all of them. The 2.5-D stage is therefore scored by its own family, defined here.

Crossing order accuracy. Each reference crossing names the instance physically in front; the reconstruction either names one or abstains. Three quantities are reported and none is quoted alone: *decided only*, the fraction of committed decisions that are correct, which by itself is not evidence because abstaining on every hard crossing scores 1.000; *all crossings*, which counts an abstention as an error and is the headline; and the *abstention rate* beside both. Chance is exactly 0.5, the decision being binary and symmetric, so only the margin above it carries information. A predicted crossing is credited only when it involves the same instance pair as a reference crossing and lies within the placement tolerance, so inventing crossings cannot help.

Depth order over instance pairs. Crossing accuracy is local. The global counterpart asks, for two instances, whether the reconstruction places the deeper one deeper, and is reported twice. Over *crossing pairs* it measures the stack where evidence exists. Over *all pairs* it includes instances that never cross and are therefore constrained by no evidence at all, whose relative depth is fixed by the solver’s tie-break rather than by the image. The two are never pooled, and the fraction of pairs sharing a component is reported beside them so the second can be read against what it could possibly measure. Both are invariant to monotone rescaling of depth, so they describe the arrangement rather than its units. Kendall’s τ over the same pairs is $2a - 1$ for accuracy a and is not reported separately.

Layer structure. Layers are an ordinal quotient of depth: a reconstruction can order every pair correctly yet cut the film into a different number of layers. Layer agreement, the fraction of instances placed in the reference layer, is therefore reported separately.

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